

Course Framework

Program Name	FSD
Course Name	Front End Frameworks
Name of Educational Developer	Yi Jin
Name of SME(s)	Karan Aggarwal
Course Hours	80
Course Delivery	Online Instructor-Guided
Course Description	Embark on a transformative journey in our Front-End Frameworks Course, where you will study the intricacies of Angular and React. First, you will examine Angular's robust architecture, mastering TypeScript and component-based development. Then, you will study React's declarative paradigm and virtual Document Object Model for efficient user interface creation. Using Microsoft Azure you will develop your skills in cloud-based development and through hands-on projects, delve into state management, routing, and industry best practices. This course will give you a comprehensive understanding of these frameworks, enabling you to craft an immersive user experience and navigate the ever-evolving landscape of web development.
Course Rationale	Students will learn to develop dynamic Single Page Web Applications using today's most popular front-end frameworks: Angular and React. Through a series of graduated assignments, students will develop an application called The Everyday Market App in each of the two frameworks. The Everyday Market App contains all the components of a standard shopping cart app and will serve as a valuable portfolio piece for students upon graduation.
Course Outcomes and Sub Outcomes	Outcome 1: Demonstrate how to build a single page application using Angular and .Net. 1. Describe the Angular framework 2. Demonstrate how to create Angular applications using the Angular CLI 3. Construct reusable modules and components 4. Demonstrate how to bind data into web elements using ngModel 5. Manage state and component interaction Outcome 2: Demonstrate how to build a single page application using React and .Net. 1. Describe the React Framework 2. Demonstrate how to create React applications using the create-react-app command

	- Construct reusable React class and functional components - Demonstrate how to use JSX to bind data and create interactive components - Demonstrate how to manage state and props in an application - Demonstrate how to coordinate interaction between React components Outcome 3: Demonstrate how to implement Routing and form validation in Angular and React - Demonstrate how to implement routing with the Angular Router - Demonstrate how to validate forms using client side validation in Angular - Demonstrate how to implement routing with the React - Demonstrate how to validate forms using client side validation in React Outcome 4: Demonstrate how to deploy web applications to the MS Azure cloud to improve the development and deployment process. - Describe cloud computing - Demonstrate how to create a Microsoft Azure account - Demonstrate how to create a Pipeline to build and deploy the code - Execute CI/CD - Continuous Integration and Continuous Deployment - Describe the basic terms: subscription, account, and ARM
Overall Assessment Scheme	Assignments 100%
Assessment Breakdown	Assignment 1: Front End Frameworks: Everyday market app - Angular- 25% Assignment 2: Everyday market app - React - 25% Assignment 3 - 25% Assignment 4: Cloud deployment with MS Azure - 25%
Required Textbook	Drescher, T., Zuker, A., & Friedman, S. (2018). Hands-On Full Stack Web Development with ASP.NET Core: Learn end-to-end web development with leading frontend frameworks such as Angular, React and Vue. Packt
Required Courseware (from Publisher)	N/A
Required Software (Not from Publisher)	

Required Equipment	Laptop/Computer
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Assessment Scheme

Assessment Title	Description	Weight
Assignment 1 - Assignment 1: Front End Frameworks: Everyday market app - Angular	Students will build single-page web applications with Angular and Implement RESTful APIs.	25%
Assignment 2 - Everyday market app - React	Students will build single-page web applications with React and Implement RESTful APIs.	25%
Assignment 3 - Implement Routing and Form validation in any one of the Assignment 1 or 2.	Students will further build their web application from Assignment 1 or 2 to implement routing and form validation.	25%
Assignment 4 - Cloud deployment with MS Azure	Students will demonstrate cloud deployment with MS Azure.	25%

Assessment Mapping

Assessment	LO1: Demonstrate how to build a single page application using Angular and .Net.	LO2: Demonstrate how to build a single page application using React and .Net.	LO3: Demonstrate how to implement Routing and form validation in Angular and React	LO4: Demonstrate how to deploy web applications to the MS Azure cloud to improve the development and deployment process.
Assignment 1	x			
Assignment 2		X		
Assignment 3			X	
Assignment 4: Cloud deployment with MS Azure				X

Module/ Topic	Course Learning Outcome and Sub-outcomes	Summative Assessments (include title, description and grade points/ weightings)	Activities (Facilitated, Peer-to-Peer & Independent) *Include HYperlinks to Textbook Chapters, Resources and Materials
Approximate time (hours) Essential Questions			
Module 1 Building Application with Angular 20 Hours	In this module, we'll delve into building a single-page application (SPA) using Angular and .NET. We'll explore the Angular framework, showcasing the creation of applications with Angular CLI	<u>Assignment1:</u> Building an Angular Application	Introductory Videos: https://drive.google.com/file/d/1tN-M7rl_yIzyMx_7cZtO8c65FcqI2Os6/view?usp=sharing Presentations:

	(Command Line Interface), building reusable modules and components, implementing data binding with ngModel, managing state and component interaction, integrating routing with the Angular Router, and validating forms through client-side validation. Outcome 1: Demonstrate how to build a single page application using Angular and .Net. LEARNING OBJECTIVES • Describe the Angular framework • Demonstrate how to create Angular applications using the Angular CLI • Construct reusable modules and components • Demonstrate how to bind data into web elements using ngModel • Manage state and component interaction • Demonstrate how to implement routing with the Angular Router		https://docs.google.com/presentation/d/1bAikCVC97PDlaapqdcix5-QQgPM57h7h/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true Videos/External Resources: https://www.youtube.com/watch?v=k5E2AVpwsko https://www.youtube.com/watch?v=9ZD7cKlaxdM https://angular.io/start Practice Activities: Setup Angular CLI and create application using Angular CLI: https://docs.google.com/document/d/12bW53OFh868FMM35ILKo5dI09LnVSUUA/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true Creating a Component: https://docs.google.com/document/d/1ktu8f7QqtNbw3wudl8fRXaGppUAdUO_l/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true
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	• Demonstrate how to validate forms using client side validation This module needs to be completed in the first week of the course and it will take approximately 20 hours. In that time, you are required to: - Read Chapter 10 – App Development with Angular - Watch the supplementary videos in the Module Readings section - Complete the practice exercises, post to the discussion forum, and provide feedback to your classmates. - Complete Assignment #1 - Make notes as you work through the information and go through the examples in the eBook. Work on implementing examples and don't hesitate to ask questions in the Q&A forum.		Routing and component Interaction : https://docs.google.com/document/d/1dOLHo_ql5fnOAK2ZV4TDVN6zxqXLSCGB/edit?usp=sharing&ouid=117675439328301646509&rtpof=true&sd=true Other Sessions/Activity: 1. Students can ask questions/clarification when learning new materials. Other students or the instructor can clarify their concerns. 2. Group project clarifications sessions: Create a group of 4 students. Each student will bring up issues they are facing while doing their assignment projects. Other students can assist guiding their colleagues but do not provide answers. 3. Go through basic angular interview questions. https://www.interviewbit.com/angular-8-interview-questions/. Further discussion in class on any concerns and interesting points that the students discover
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Module 2 Building Application with React 20 Hours	Learn how to create a single page application with React and .NET. Uncover the essence of the React framework, learn to craft applications using the create-react-app command, build reusable class and functional components, utilize JSX for data binding, and manage state, props, and component interaction. Outcome 2: Demonstrate how to build a single page application using React and .Net. LEARNING OBJECTIVES: • Describe the React Framework • Demonstrate how to create React applications using the create-react-app command • Construct reusable React class and functional components • Demonstrate how to use JSX to bind data and create interactive components • Demonstrate how to manage state and props in an application	Assignment - 2 Building a React Application: https://docs.google.com/document/d/1zx0eaxsugLIGWyfEOTki55EmHACBdAco/edit?usp=sharing&ouid=117675439328301646509&rtpof=true&sd=true	Presentations: https://docs.google.com/presentation/d/1uJSFvILADbgI579hBtxrvECAOarnRPo/edit?usp=sharing&ouid=117675439328301646509&rtpof=true&sd=true Videos/External Resources: https://www.youtube.com/watch?v=SqcY0GIETPk https://www.youtube.com/watch?v=4RKuyp_bOhY https://legacy.reactjs.org/docs/getting-started.html#learn-react Practice Activities: Create React applications using the create-react-app command: https://docs.google.com/document/d/1NFpSb3FAHTqfmsAiAMKuU6ELe6PvPjZe/edit?usp=sharing&ouid=117675439328301646509&rtpof=true&sd=true Creating a Component: https://docs.google.com/document/d/1awA-d7i7VAu6D8H2_rMfhDwvmf4H0VVK/edit?usp=sharing&ouid=117675439328301646509&rtpof=true&sd=true
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	• Demonstrate how to coordinate interaction between React components This module needs to be completed in the second week of the course and it will take approximately 20 hours. In that time, you are required to: - Read Chapter 12 – App Development with React - Watch the supplementary videos in the Module Readings section - Complete the practice exercises, post to the discussion forum, and provide feedback to your classmates. - Complete Assignment #2 - Make notes as you work through the information and go through the examples in the eBook. Work on implementing examples and don't hesitate to ask questions in the Q&A forum.		developing a Homepage: https://docs.google.com/document/d/1RyHdNeHM7SQpAb4DX2GWnYd82Vm8tSU4/edit?usp=sharing&ouid=117675439328301646509&rtpof=true&sd=true Other Sessions/Activity: 1. Students can ask questions/clarification when learning new materials. Other students or the instructor can clarify their concerns. 2. Group project clarifications sessions: Create a group of 4 students. Each student will bring up issues they are facing while doing their assignment projects. Other students can assist guiding their colleagues but do not provide answers. 3. Go through React interview questions.. https://www.interviewbit.com/react-interview-questions/. Further discussion in class on any concerns and interesting points that the students have discovered.
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Module 3 Routing and Form validation in React and Angular. 20 Hours	Learn Angular and React essentials with practical lessons! Explore Angular Router to create dynamic web navigation and learn client-side form validation techniques for a seamless user experience. Examine React's routing system for efficient single-page applications and ensure data integrity with client-side form validation in React. Outcome 3: Demonstrate how to implement Routing and form validation in Angular and React LEARNING OBJECTIVES • Demonstrate how to implement routing with the Angular Router • Demonstrate how to validate forms using client side validation • Demonstrate how to implement routing with the React • Demonstrate how to validate forms using client side validation in React	Assignment - 3: Implementing Routing and form validation https://docs.google.com/document/d/1wod4U_GLE0t3rmS93t_n069gRXEYzWE-/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true	Presentations: https://docs.google.com/presentation/d/10bFtGBVuiWwkxREzOD73bb_-NqmiyOrt/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true Videos/External Resources: https://www.youtube.com/watch?v=kaHSVI4kDTc https://www.youtube.com/watch?v=a2fGh8wqHmo https://www.youtube.com/watch?v=TWz4TjSssbq https://www.youtube.com/watch?v=tIdNeoHniEY Practice Activities: Form Validation in Angular: https://docs.google.com/document/d/19ly9--JnsGudeZiVzzdyA-UhmWSR9njb/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true Form Validation in React: https://docs.google.com/document/d/15YoWLknIM76VH7XOWOXMaHmb9IBDPssy/
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	This module needs to be completed in the third week of the course and it will take approximately 20 hours. In that time, you are required to: - Read Chapter 11 – Implementing Routing and Forms - Watch the supplementary videos in the Module Readings section - Complete the practice exercises, post to the discussion forum, and provide feedback to your classmates. - Complete Assignment #3 - Make notes as you work through the information and go through the examples in the eBook. Work on implementing examples and don't hesitate to ask questions in the Q&A forum.		edit?usp=sharing&ouid=117675439328301646509&rtpof=true&sd=true Other Sessions/Activity: 1. Students can ask questions/clarification when learning new materials. Other students or the instructor can clarify their concerns. 2. Group project clarifications sessions: Create a group of 4 students. Each student will bring up issues they are facing while doing their assignment projects. Other students can assist guiding their colleagues but do not provide answers. Debate Session: Divide the class in two groups. Assign Angular to one while React to another. Students debate why their framework is better and why other frameworks should not be used.
Module 4 Web Application deployment in Microsoft Azure 20 Hours	Unlock the power of cloud computing by examining the deployment of web applications on the Microsoft Azure platform. From understanding cloud computing	Assignment - 4 Deploying app on Microsoft Azure:	Presentations: https://docs.google.com/presentation/d/1J075GS5_eQkCHzE8XqjScGdwfc9BbzVq/edit?usp=sharing&ouid=117675439328301646509&rtpof=true&sd=true

	principles to creating an Azure account, setting up CI/CD (Continuous Integration/Continuous Deployment) pipelines, and understanding terms like subscription, account, and ARM (Azure Resource Manager), this module will enable you to streamline development and deployment processes. Outcome 4: Demonstrate how to deploy web applications to the MS Azure cloud to improve the development and deployment process. LEARNING OBJECTIVES • Describe cloud computing • Demonstrate how to create a Microsoft Azure account • Demonstrate how to create a pipeline to build and deploy the code • Execute CI/CD - Continuous Integration and Continuous Deployment • Describe the basic terms: subscription, account, and ARM	https://docs.google.com/document/d/1X92uyz78sMDMy09BC0FW0QtAhPh2Tqv3/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true	Videos/External Resources: https://www.youtube.com/watch?v=LwDYfL7mGfw Practice Activities: Microsoft Azure, pipeline and code deploy: https://docs.google.com/document/d/1J4Xn8vld_Hm1uHaz8SyojX2kUDbbqRbP/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true Continuous Integration and Continuous Deployment: https://docs.google.com/document/d/1rRSxu7UHL6vbHIGdQ8nsuRwL_kkXI5I/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true Execute CI/CD: https://docs.google.com/document/d/1B20KAOhrRJOLINCHCh8wEIPmhbuWw_MfT/edit?usp=sharing&oid=117675439328301646509&rtpof=true&sd=true
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	This module needs to be completed in the fourth week of the course and it will take approximately 20 hours. In that time, you are required to: - Read Chapter 14 – Cloud Deployment - Watch the supplementary videos in the Module Readings section - Complete the practice exercises, post to the discussion forum, and provide feedback to your classmates - Complete Assignment #4 - Make notes as you work through the information and go through the examples in the eBook. Work on implementing examples and don't hesitate to ask questions in the Q&A forum.		Other Sessions/Activity: 1. Students can ask questions/clarification on concerns in their learning. Other students or the instructor can clarify their concerns. 2. Group project clarifications sessions: Create a group of 4 students. Each student will bring up issues they are facing while doing their assignment projects. Other students can assist guiding their colleagues but do not provide answers. 3. Drawing on what students have learned in the course, describe, in a discussion group posting, an innovative application that could be developed using Angular/React and Cloud. Discuss what resources would be needed and how the application could work in the actual market.
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